

ARYAN SEHGAL

SOFTWARE ENGINEER — FRONTEND & APPLIED AI/ML

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PROFESSIONAL PROFILE

Software engineer with 2+ years of experience building high-performance, accessible web products, and a B.Tech in Computer Science & Engineering (Artificial Intelligence). At Sprinklr, worked as one of two core engineers building the Ads Creative Management application end-to-end — a lean-team environment that, alongside primary frontend ownership, gave direct exposure to backend services and AI-agent API integrations behind the product's AI-assisted brand-guideline generation and creative-scoring features. Since leaving Sprinklr, has pursued self-directed, in-depth AI/ML study — workshops, primary research papers, and reproducible public projects spanning generative modelling, on-device NLP, and classical machine learning — including training a custom GAN from published research through to a deployed inference service. Seeking an MSc in Artificial Intelligence to build rigorous theoretical and research foundations alongside this applied engineering experience.

EDUCATION

Netaji Subhas University of Technology Jul 2019 – May 2023

B.Tech, Computer Science & Engineering (Artificial Intelligence) | CGPA: 8.69 / 10

Class XII, Central Board of Secondary Education (CBSE) — Presidium, New Delhi | 90% | 2019

Class X (CBSE) — DAV Centenary Public School, New Delhi | CGPA: 10/10 | 2017

PROFESSIONAL EXPERIENCE

Independent Software Engineer & ML Developer | *Freelance Projects & Continuing Study* Sep 2025 – Present

- Design, build, and deploy end-to-end web and AI applications; maintain a public GitHub portfolio of product prototypes and reproducible Jupyter case studies.
- Deepen machine-learning foundations through workshops, tutorials, and primary research papers; apply the learning to generative modelling, on-device inference, and classical ML techniques across the projects below.

Senior Product Engineer | *Sprinklr* Jul 2023 – Sep 2025

- One of two core engineers building Sprinklr's Ads Creative Management application from an early stage; owned frontend architecture end-to-end while also engaging with backend services and the AI-agent API integrations behind AI-assisted brand-guideline generation and creative-scoring capabilities.
- Revamped the Digital Asset Manager with server-side rendering, lazy loading, and code splitting, cutting page-load time by 35%; raised frontend test coverage to 95% with Jest and React Testing Library.
- Modernized a legacy React codebase from class-based to functional components and migrated client data management from Redux/REST to GraphQL with Apollo Client, reducing API response times by 40%.
- Improved the mobile responsiveness and accessibility (WCAG) of the in-house Spacweb Editor while reducing its bundle size by 25%; built a reusable Progress Steps component for the internal Hyperspace design system.

Data Science Intern | *OYO Rooms* May 2022 – Jul 2022

- Cleaned, filtered, and preprocessed data; trained ML models for predictive analytics and built a web interface for data visualization.

Teaching Assistant | *Coding Ninjas* Aug 2020 – Nov 2020

- Mentored students through data structures, algorithms, and competitive-programming problem solving.

SELECTED APPLIED AI & SOFTWARE PROJECTS

Public portfolio: github.com/AryanSehgal

Style-Guided Face-to-Anime Translation (StyleFAT) — Research to Production | Python, TensorFlow/Keras, PyTorch, tf2onnx, ONNX Runtime, Node.js, React, TypeScript

- Independent generative-AI research project: studied GAN and image-to-image translation literature (CycleGAN-adjacent work, DRIT++, EGSC-IT, AniGAN, StarGAN), then designed, implemented, and trained a custom GAN from scratch to translate photographic portraits into anime-style faces.
- Converted the trained model (HDF5) to ONNX with tf2onnx and integrated it into a full-stack web application (React/TypeScript frontend, Node.js/Express backend, ONNX Runtime CPU inference) with interactive before/after comparison. [Live demo](#)

Edge AI Screen & Audio Note Studio | *React 19, TypeScript, Vite, Transformers.js, ONNX Runtime Web, Web Workers, IndexedDB*

- Built a 100% client-side note-taking studio: on-device Whisper (ONNX, WebGPU with WASM fallback) transcribes recorded or uploaded audio, and an on-device FLAN-T5 model drafts titles, summaries, and action items — zero external AI API calls, zero backend, Lighthouse Performance score of 100.
- Live: [edge-ai-note-studio.vercel.app](#)

Chroma — Black-and-White Image Colorization | *React, TypeScript, FastAPI, OpenCV (pretrained model)*

- Full-stack photo studio that applies a pretrained deep-learning colorization model on a dedicated backend; supports batch upload, adjustable color intensity, before/after comparison, and individual or ZIP downloads. [Live demo](#)

Forma UI & Forma Visual Page Builder | *React 19, TypeScript, Radix primitives, Tailwind CSS, npm*

- Authored and published Forma UI, a 16-component accessible React design-system package (@aryan_sehgal/forma-ui) with semantic theming tokens, an interactive documentation site, and a CI/CD pipeline that auto-publishes releases to npm. [Playground](#)
- Used the library to build a drag-and-drop visual landing-page builder — 10 content blocks, responsive desktop/tablet/mobile artboards, undo/redo history, and static-site export. [Live demo](#)

Machine Learning Case Studies (Self-Study) | *Python, pandas, scikit-learn, XGBoost, LightGBM, statsmodels, Jupyter*

- IPL Match Analytics (2008–2024): cleaned and feature-engineered 1,095 matches across 20 features; built K-Means team-performance clustering, logistic regression, KNN, and a decision-tree toss-strategy model, then compared Random Forest, AdaBoost, XGBoost, and a Voting Ensemble for match-outcome prediction — finding that the toss itself had virtually no bearing on the result (won by only 50.6% of toss winners).
- E-commerce Marketing & Sales Analysis: merged five transactional and marketing data sources into a unified dataset; performed RFM-based and K-Means customer segmentation, cohort/retention analysis, and marketing-spend correlation analysis, translating findings into strategic recommendations on personalization, retention, and channel investment.
- Additional case studies applying the classical ML toolkit end-to-end: vehicle CO₂-emissions prediction (linear/polynomial regression, VIF-based multicollinearity reduction, permutation importance; $R^2 \approx 0.98$); time-series forecasting (ARIMA/SARIMA/SARIMAX, exponential smoothing, Prophet); clustering (K-Means, Hierarchical, GMM, DBSCAN); anomaly detection (IQR, Elliptical Envelope, Isolation Forest, LOF); ensemble learning (Random Forest, GBDT, XGBoost, LightGBM); Naive Bayes text classification; and recommender systems (collaborative filtering & SGD-based matrix factorization).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, C++, HTML5/CSS3

AI, ML & Data: TensorFlow/Keras, PyTorch, ONNX & ONNX Runtime, Transformers.js, OpenCV, scikit-learn, XGBoost, LightGBM, statsmodels, NumPy, pandas, Jupyter

Frontend & Product Engineering: React, Next.js, Redux/Redux Toolkit, GraphQL, Apollo Client, Tailwind CSS, Radix primitives, WCAG/ARIA accessibility, responsive design

Backend, Testing & Delivery: Node.js, Express, FastAPI, REST APIs, Web Workers, IndexedDB, Jest, React Testing Library, Git, GitHub Actions, Vercel, Netlify

CERTIFICATIONS

- Front-End System Design — Frontend Masters (Sep 2025)
- Build an Epic React App — Kent C. Dodds, EpicReact.dev (Jul 2023)
- CSS for JavaScript Developers — Josh W. Comeau (Jul 2023)
- Certificate of Excellence, Data Structures in C++ — Coding Ninjas
- Certificate of Excellence, Python Programming — Coding Ninjas

ACADEMIC ACHIEVEMENTS

- All India Rank 1,912 in JEE Main and All India Rank 5,021 in JEE Advanced.
- Solved 750+ problems across competitive-programming platforms; earned 16 LeetCode badges.